BSDS: III: Regression Techniques
First Course Handout

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Lecture hours: Tuesday and Wednesday 10:30 AM - 12 noon

Tutorial: Thursday 12:05 – 1:00 PM

Course webpage:

https://sites.google.com/view/mmukhopadhyay/home/teaching/regression-techniques?authuser=0

Objective:

This course aims to provide students with a comprehensive understanding of multiple linear regression, with emphasis on both the theoretical foundations and practical aspects of statistical modeling. Students will develop the ability to interpret regression models geometrically, perform estimation, prediction, and statistical inference, diagnose violations of model assumptions such as outliers, heteroscedasticity, autocorrelation, and multicollinearity, and apply appropriate remedial measures. The course also introduces systematic approaches to model building and selection, including variable selection techniques and model evaluation criteria, enabling students to construct, assess, and interpret reliable regression models for data analysis and decision-making.

References:

1. Seber, G. A. F., & Lee, A. J. (2003). *Linear Regression Analysis* (2nd ed.). John Wiley & Sons. [Textbook, Link]
2. Montgomery, D. C., Peck, E. A., & Vining, G. G. (2021). *Introduction to Linear Regression Analysis* (6th ed.). John Wiley & Sons. [Link]
3. Draper, N. R., & Smith, H. (1998). *Applied Regression Analysis* (3rd ed.). John Wiley & Sons. [Link]
4. Ryan, T. P. (2008). *Modern Regression Methods* (2nd ed.). John Wiley & Sons.
5. Sengupta, D., & Jammalamadaka, S. R. (2003). *Linear Models: An Integrated Approach*. World Scientific Publishing. [Link]
6. Fox, J. (2016). *Applied Regression Analysis and Generalized Linear Models* (3rd ed.). SAGE Publications.

Course Content: We will cover the following contents in this course:

1. Multiple linear regression: Properties of least squares estimators and residuals.
2. Geometric view of regression:
    a. Orthogonal projections on subspaces associated with predictors.
    b. Decomposition of sum of squares. Degrees of freedom.
    c. Multiple and partial correlation coefficients.
3. Prediction in linear regression: Point and interval prediction. Measures of prediction accuracy.
4. Inference on linear regression:
    a. Confidence and prediction intervals.
    b. Tests of hypotheses for linear parametric functions.
    c. Simultaneous inference.
5. Violation of linear model assumptions:
    a. Outliers: high-leverage and influential observations, consequences, diagnostics.
    b. Heteroscedasticity: consequences, diagnostics, tests (Bruesch-Pagan and White's test).
    c. Autocorrelated errors: consequences, diagnostics, ACF and PACF plots, tests (Durbin-Watson test).
6. Methods of addressing model violation:
    a. Variable transformation (Box-Cox family).
    b. Weighted least squares.
7. Multicollinearity: Consequences, diagnostics. Notion of variable importance.
8. Model building:
    a. Subset selection, Forward and backward stepwise regression.
    b. Categorical predictors and dummy variables.
9. Model selection:
    a. Prediction and model error.
    b. Adjusted R2, FPE, Mallows' Cp, AIC, BIC criteria.
    c. Cross-validation approach.

Grading and Evaluation:

(i) There will be 2 **quizzes**, a mid-semester and an end-semester exams.

(ii) There will be a group project carrying 10% weightage included in the endsem.

(iii) Marking scheme is *absolute*. The weights of different evaluations are as follows:

Quizzes: 20% on best of two quizzes

Midterm: 30%

Endterm: 50% (including group project carrying 10% weightage)

(iv) If a student misses a quiz, then s/he will be evaluated based on the other quiz. If s/he misses both the quizzes, then s/he will get *zero* marks in quiz.

(v) The answer scripts after evaluation will be uploaded in the GradeScope.

General Guidelines:

(i) Students are expected to spend 10 - 12 hours per week for good performance in this course.

(ii) If a student found to follow unfair means (in any of the exams), then irrespective of the circumstances his/her exam will be cancelled (with no supplementary exam).

Project Guidelines:

(i) A group projects could either be a data analysis project, or self-study of a topic not covered in class

(ii) The students are requested to form groups of size at most three. The deadline for group registration is 31^{st} July, 2026.

(iii) Deadline to decide type of project with a tentative topic and/or data description: 31^{st} August, 2026.

(iv) Deliverables include a project report and presentation slides, to be submitted by 6^{th} November, 2026.

(v) Presentations will be scheduled during the last two weeks of class, or during the study leave.